

NAME: Shruthe

ROLL NO.: 241801265

Back To Practice

Non-Preemptive Shortest Job First CPU Scheduling Algorithm

Write a C program to implement the Non-Preemptive Shortest Job First (SJF) CPU Scheduling Algorithm. The program should accept the number of processes, their Arrival Times (AT) and Burst Times (BT). At any given time, the process with the minimum burst time among the arrived processes must be executed first. Calculate the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process and find the overall average TAT and WT.

Core Logic

Selection: Out of all processes that have arrived by the current time, the one with the minimum Burst Time is chosen.

Formulas:

Select Language: C (GCC 9.2.0)

```
1 #include <stdio.h>
2 int main() {
3     int n, i, j, time = 0, completed = 0;
4     float avgTAT = 0, avgWT = 0;
5
6     printf("Enter number of processes: ");
7     scanf("%d", &n);
8
9     int AT[n], BT[n], CT[n], TAT[n], WT[n], done[n];
10
11     for(i = 0; i < n; i++) {
12         printf("Enter Arrival Time and burst time for PID: ", i+1);
13         scanf("%d %d", &AT[i], &BT[i]);
14         done[i] = 0;
15     }
16
17     while(completed < n) {
18         int idx = -1;
```

Select Language: C (GCC 9.2.0)

```
31         time += BT[idx];
32         CT[idx] = time;
33         TAT[idx] = CT[idx] - AT[idx];
34         WT[idx] = TAT[idx] - BT[idx];
35
36         avgTAT += TAT[idx];
37         avgWT += WT[idx];
38
39         done[idx] = 1;
40         completed++;
41     }
42     else {
43         time++;
44     }
45 }
46
47 avgTAT /= n;
48 avgWT /= n;
49
50 printf("\n\nAverage Turnaround Time = %.2f\n", avgTAT);
51 printf("Average Waiting Time = %.2f\n", avgWT);
52
53 return 0;
54 }
```

Test Cases 2

Run Sample Cases

Run All Cases

Custom test case

Custom Test



Input:

4
0 6
0 8
0 7
0 3

Expected Output:

Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:
Enter Arrival Time and Burst Time for P4:

Average Turnaround Time = 13.00
Average Waiting Time = 7.00

Output:

Enter number of processes:
Enter Arrival Time and Burst Time for P1:
Enter Arrival Time and Burst Time for P2:
Enter Arrival Time and Burst Time for P3:
Enter Arrival Time and Burst Time for P4:

Average Turnaround Time = 13.00
Average Waiting Time = 7.00